Lecture 20: Polar Coordinates

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

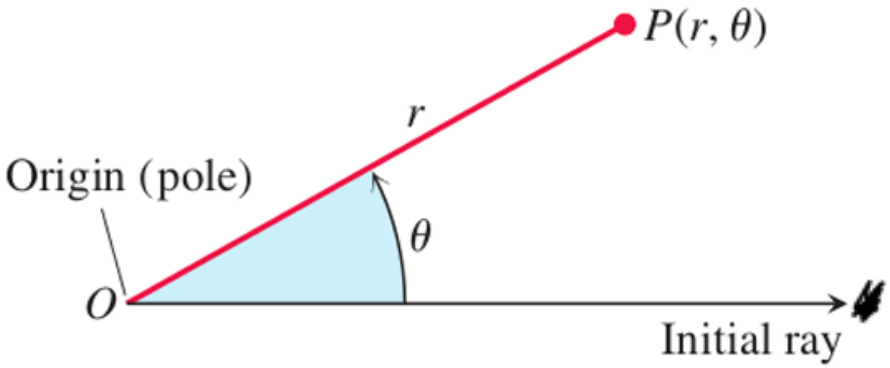
Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand the definition of polar coordinates.
- ☐ Identify the geometric meaning of polar equations and recognize common polar graphs.
- ☐ Sketch graphs of polar equations.
- ☐ Convert equations between polar and Cartesian coordinate systems.

Polar Coordinates

Many problems in mathematics are easier to solve if we choose a suitable coordinate system. Previously, we use Cartesian coordinate. In this section, we study polar coordinates.



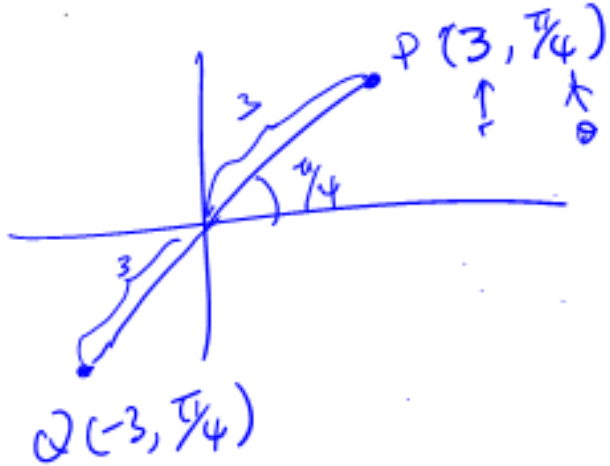
Fix an origin O (pole) and an initial ray from O. Then each point P can be located by assigning it to a polar coordinate pair (r, θ) where

$$r = \text{ the directed distance from } O \text{ to } P,$$
$$\theta = \text{ the directed angle from the initial ray to ray } OP.$$

- θ positive when we measure the angle counterclockwise.

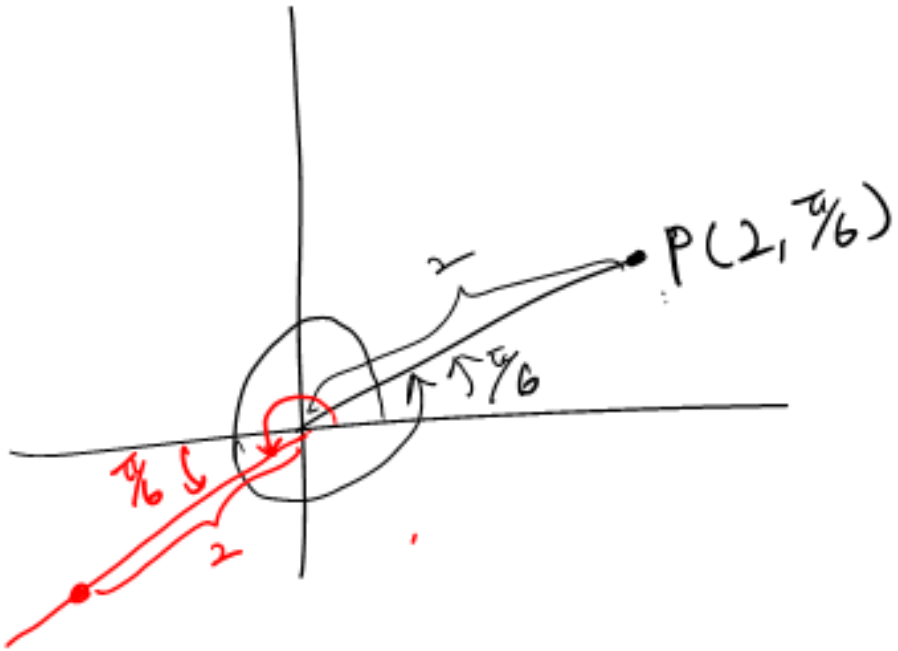
  θ negative when we measure the angle clockwise.

- r negative - the ray is going backward (in opposite direction)



Polar Coordinates

Example: Find all the polar coordinates of the point $P(2, \pi/6)$.



$P(2, \pi/6) = P(2, \pi/6 + 2n\pi)$, $n \in \mathbb{Z}$.

$= P(-2, \frac{7\pi}{6} + 2n\pi)$, $n \in \mathbb{Z}$

Eg: $P(2, \frac{\pi}{6}) = P(2, \frac{13\pi}{6})$

$= P(-2, \frac{7\pi}{6})$

$= P(-2, \frac{12\pi}{6})$

Polar Equations and Graphs

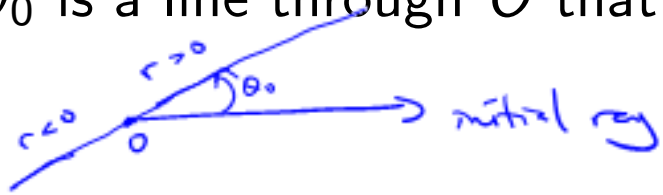
Previously, in Cartesian coordinate system. $x = c$ is a vertical $y = d$ is a horizontal

In the polar coordinate system, the equations

(i) $r = a$ is a circle with radius $|a|$ centered at O, and



(ii) $\theta = \theta_0$ is a line through O that makes an angle θ_0 with the initial ray.



Example:

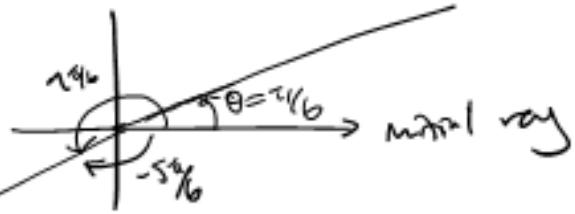
Taking all values of θ

(i) $r = 1$ and $r = -1$ are equations for the circle of radius 1 centered at O.



Taking all values of r

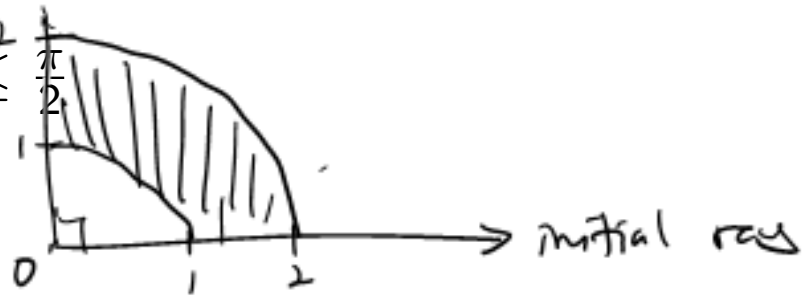
(ii) $\theta = \pi/6$, $\theta = 7\pi/6$ and $\theta = -5\pi/6$ are equations for a line.



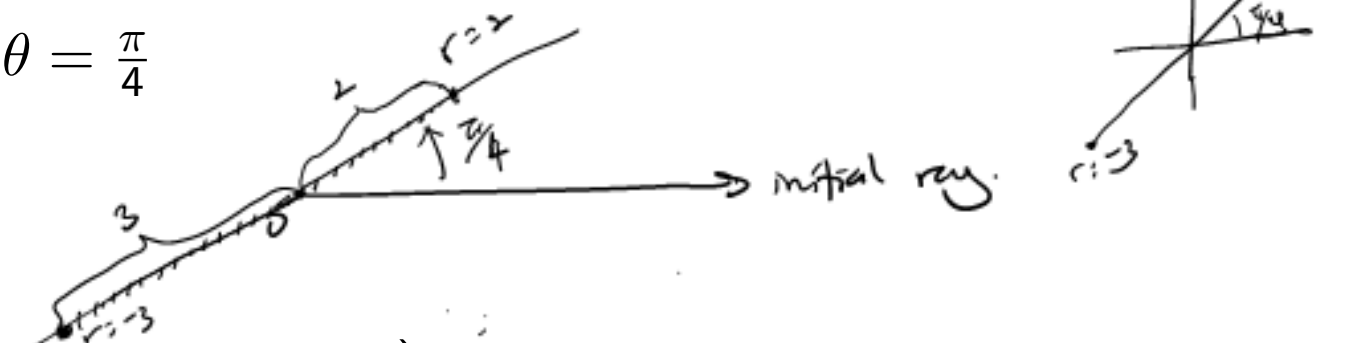
Polar Equations and Graph

Example: Graph the sets of points whose polar coordinates satisfy the following conditions.

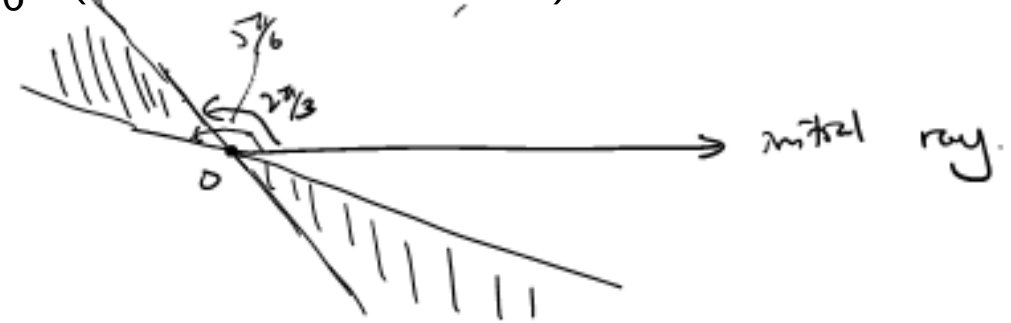
(a) $1 \leq r \leq 2$ and $0 \leq \theta \leq \frac{\pi}{2}$



(b) $-3 \leq r \leq 2$ and $\theta = \frac{\pi}{4}$



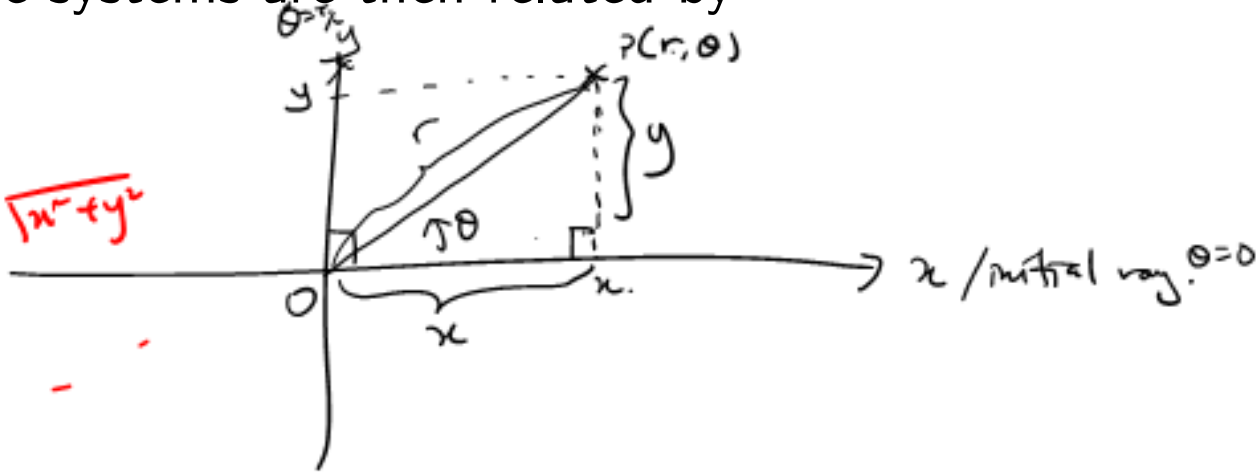
(c) $\frac{2\pi}{3} \leq \theta \leq \frac{5\pi}{6}$ (no restriction on r)



Relating Polar and Cartesian Coordinates

In both systems, if we place the two origins together, then the initial polar ray be the positive x-axis and the ray $\theta = \frac{\pi}{2}$, $r > 0$ becomes the positive y-axis. The two coordinate systems are then related by

- $x = r\cos\theta$;
- $y = r\sin\theta$;
- $r^2 = x^2 + y^2$; $\Rightarrow r = \sqrt{x^2+y^2}$
- $\tan\theta = \frac{y}{x}$.



Example:

$r\cos\theta = 2 \Rightarrow x = 2$

$r^2\cos\theta\sin\theta = 4 \Rightarrow (r\cos\theta)(r\sin\theta) = 4 \Rightarrow xy = 4$

$r^2\cos^2\theta - r^2\sin^2\theta = 1 \Rightarrow (r\cos\theta)^2 - (r\sin\theta)^2 = 1 \Rightarrow x^2 - y^2 = 1$

$r = 1 + 2r\cos\theta \Rightarrow \sqrt{x^2+y^2} = 1 + 2x \Rightarrow x^2+y^2 = 1 + 4x + 4x^2 \Rightarrow y^2 - 3x^2 - 4x - 1 = 0$

$r = 1 - \cos\theta \Rightarrow r^2 = r - r\cos\theta \Rightarrow x^2+y^2 = \sqrt{x^2+y^2} - x \Rightarrow x^4 + y^4 + 2x^2y^2 + 2x^3 + 2xy^2 - y^2 =$

Relating Polar and Cartesian Coordinates

Find a polar equation for the circle $x^2 + (y-3)^2 = 9$. $= 3^2$

circle center at $(0, 3)$ with radius 3

$$x^2 + (y-3)^2 = 9$$

$$(r\cos\theta)^2 + (r\sin\theta - 3)^2 = 9$$

$$r^2\cos^2\theta + r^2\sin^2\theta - 6r\sin\theta + 9 = 9$$

$$r^2(\cos^2\theta + \sin^2\theta) - 6r\sin\theta = 0$$

$$r^2 - 6r\sin\theta = 0$$

$$r(r - 6\sin\theta) = 0$$

$r = 0$ or $r - 6\sin\theta = 0$

X definitely r 3 not only 0

So the polar equation is $r = 6\sin\theta$.

Relating Polar and Cartesian Coordinates

Replace the following polar equations by equivalent Cartesian equations and identify their graphs.

(a) $r\cos\theta = -4$

(b) $r^2 = 4r\cos\theta$

(c) $r = \frac{4}{2\cos\theta - \sin\theta}$

(d) $r = 2\cos\theta$

(a) $x = -4$ A vertical line with $x = -4$.

(b) $x^2 + y^2 = 4x$

$x^2 - 4x + y^2 = 0$

$(x-2)^2 - 4 + y^2 = 0$

$(x-2)^2 + y^2 = 4$

A circle of radius 2 with center $(2, 0)$

(c) $r = \dfrac{4}{2\cos\theta - \sin\theta}$

$2r\cos\theta - r\sin\theta = 4$

$2x - y = 4$

$y = 2x - 4$

A straight line with gradient 2 and y-intercept -4.

(d) $r = 2\cos\theta$.

$r^2 = 2r\cos\theta$

$x^2 + y^2 = 2x$

$x^2 - 2x + y^2 = 0$

$(x-1)^2 - 1 + y^2 = 0$

$(x-1)^2 + y^2 = 1$

A circle with radius 1 and center at $(1,0)$